

8. Brachytherapy is a form of radiotherapy used in treating prostate cancer.
Small pellets containing radioactive material are planted inside the prostate.

(a) One patient has 75 pellets containing iodine-125 planted in his prostate.
Each pellet has an initial activity of 16 MBq.

(i) Calculate the initial activity of **all** 75 pellets. [1]

initial activity = MBq

(ii) The iodine-125 has a half-life of 60 days.
This means that the activity halves every 60 days.
Calculate the activity of the 75 pellets after 120 days. [2]

activity after 120 days = MBq

(b) Iodine-123 is another isotope of iodine.
It has a half-life of 13 hours.
Explain why iodine-123 is not suitable for this treatment. [2]

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